Critical Angle, Total Internal Reflection, and Brewster Angle

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Lecture Objective

By the end of this lecture, you will be prepared to:

- Describe the concepts of critical angle, total internal reflection, and Brewster angle.
- Compute critical and Brewster angles for the interface between two dielectric half-spaces.

Critical Angle (1 of 2)

- Recall Snell's law of refraction: $$\sin\theta_t = \frac{n_1}{n_2}\sin\theta_i$$
- Case 1: $n_1 < n_2 \;\rightarrow\; \theta_t < \theta_i$
  - *'When propagating from an optically less dense medium to an optically more dense medium, the ray representing the propagation of the plane wave will bend toward the normal.'*
- Case 2: $n_1 > n_2 \;\rightarrow\; \theta_t > \theta_i$
  - *'When propagating from an optically more dense medium to an optically less dense medium, the ray representing the propagation of the plane wave will bend away from the normal.'*

Critical Angle (2 of 2)

- For case 2, there will be a ***critical angle*** of incidence (θ_c) for which the angle of refraction (θ_t) is 90 degrees.
- For $\theta_i > \theta_c$

$$\sin\theta_t > 1$$

$$\cos\theta_t = \sqrt{1-\sin^2\theta_t} = -j\sqrt{\sin^2\theta_t - 1}$$

$$e^{-jk_2 z\cos\theta_t} = e^{-k_2 z\sqrt{\sin^2\theta_t - 1}}$$ Evanescent wave

$$e^{-jk_2 x\sin\theta_t} \Rightarrow v_{px} = \frac{\omega}{k_2\sin\theta_t} < \frac{\omega}{k_2}$$ Slow wave

$$\left|\Gamma_{\perp,\parallel}\right| = 1$$

Question

- Consider a plane wave traveling in a non-magnetic dielectric medium with dielectric constant of 4 arriving at a planar interface with free space at an angle of 60 degrees wrt the normal.
- (a) Compute the critical angle of incidence for this interface. Note that it is less than 60 degrees.
- (b) Write expressions for the propagation factors in the x- and z-directions for the transmitted wave.

Total Internal Reflection

- When the angle of incidence is greater than the critical angle, the incident wave will experience total internal reflection in medium 1 and will be accompanied by a surface wave in medium 2 which propagates along the boundary.
- The field in medium 2 is <u>not zero</u>, but there is no real power propagating away from the boundary in medium 2.
- This phenomenon occurs for both perpendicular and parallel polarizations.

Brewster Angle (1 of 2)

- The Brewster angle of no reflection is the angle at which the reflection coefficient is zero.
- This phenomenon is polarization dependent.
- For perpendicular polarization, the Brewster angle occurs for

$$\theta_{B\perp} = \mathrm{a}\sin\left(\sqrt{\frac{1-\frac{\mu_1\varepsilon_2}{\mu_2\varepsilon_1}}{1-\left(\frac{\mu_1}{\mu_2}\right)^2}}\right)$$

No Brewster angle for perpendicular polarization exists for non-magnetic media.

Brewster Angle (2 of 2)

- For parallel polarization, the Brewster angle occurs for

$$\theta_{B\parallel} = \mathrm{a}\sin\left(\sqrt{\frac{1-\frac{\mu_2\varepsilon_1}{\mu_1\varepsilon_2}}{1-\left(\frac{\varepsilon_1}{\varepsilon_2}\right)^2}}\right)$$

- For non-magnetic media, we have

$$\theta_{B\parallel} = \mathrm{a}\sin\left(\frac{1}{\sqrt{1+\left(\frac{\varepsilon_1}{\varepsilon_2}\right)}}\right)$$

The Brewster angle for parallel polarization always exists for non-magnetic media.

Summary

- Snell's law of refraction reveals the existence of a critical angle when a plane wave is obliquely incident on the boundary with a less optically dense medium.
- For angles of incidence greater than the critical angle, total internal reflection occurs, and no power is transmitted across the interface.
- At the boundary between two non-magnetic media, a Brewster angle exists for parallel polarization for which the reflection coefficient is zero.

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